

Telcofy

Human Mobility Science — Powered by Telco Data

Real-Time Mobility Monitoring: Case Studies — Field Observations 2025

Ullevaal Stadion — Norway national-team match days (Sep–Oct 2025) and Tøyenparken Large Open-Air Festival (Aug 2025)

This report presents a series of real-world deployments of Telcofy's Human Mobility Science platform, leveraging anonymised and aggregated mobile network signals from Telenor to derive actionable insights on crowd density, visitor behaviour, and event analytics. All data is processed in compliance with applicable privacy regulations.

Executive Summary

This report combines two complementary mobility case studies delivered by Telcofy in 2025. The first is the Tøyenparken Large Open-Air Festival (6–9 August 2025), where Telcofy applied a 45% Telenor market-share extrapolation to monitor an open public park hosting major pop headliners. The second is a deep-dive into three consecutive Norway national-team fixtures at Ullevaal Stadion: Norway–Finland (4 September 2025), Norway–Moldova (9 September 2025), and Norway–Israel (11 October 2025). Because Ullevaal's stadium-grade cellular infrastructure is dimensioned to serve match attendees, the device counts within the stadium polygon correlate closely with official ticketing figures, providing a high-confidence reference point for crowd analytics.

Key finding — The Norway–Moldova fixture on 9 September 2025 produced the highest concurrent crowd of the entire monitoring window, with the Ullevaal Stadion polygon peaking at 29,811 people at 20:00 local time. This fully matches the stadium's 28,000 effective capacity plus surrounding ambient population. The 11–1 victory featuring five goals from Erling Haaland turned what would normally be a low-prestige opponent into the most attended fixture of the autumn window — confirming the "Haaland Effect" already observed in primary and secondary ticket markets.

International presence — Across all three matches, the share of foreign devices inside the stadium polygon ranged from 10.6% (Moldova) to 13.8% (Finland), with Germany, the Netherlands, Ukraine, Poland and Denmark consistently appearing as the top non-Norwegian source markets. Notably, the Norway–Israel fixture saw Ukraine rise to the third-largest foreign contingent (24,041 device readings), reflecting Norway's resident Ukrainian community attending what international media described as a politically charged event. Israeli SIMs registered only 2,554 readings on match day — consistent with the very small contingent of travelling supporters reported by Norwegian authorities.

Large Open-Air Festival — Tøyenparken

6–9 August 2025

By tracking Telenor network signals across the Tøyenparken geofenced zone and applying a 45% market-share extrapolation, Telcofy established a baseline for park density and visitor behaviour across the four-day event.

22,000

Average evening crowd

Peak capacity: 30,000 during headlining slots

Attendance was heavily front-loaded toward the beginning of the week:

- **August 6 (Day 1):** Chappell Roan drew the largest concentrated crowd, peaking between 26,000 and 28,000 visitors from 18:00 to 20:00.
- **August 7 (Day 2):** High density continued during Charli XCX's set, maintaining peak levels established on Day 1.
- **August 8–9:** Data shows a significant drop in visitor volume for the final two days compared to the opening pop-focused performances.

Analytics identified approximately 1,500 international devices, primarily originating from the United Kingdom and the United States. While this indicates growing international interest, the raw data also includes roaming IoT devices and automated hardware that may slightly over-represent the individual visitor count.

Temporary Antenna Dependency

Monitoring an event of this scale would have been impossible without the temporary antennas Telenor installed just before the festival — the permanent mobile network around Tøyenparken is not designed for the simultaneous load created by tens of thousands of devices in a few hectares of open park.

Temporary antenna data was partially missing on August 6th — immediately visible in the real-time dashboard as abnormally low visitor counts. Telcofy alerted Telenor, who added the missing antennas to the data feed. Data was subsequently restored using aggregated network performance records, enabling a complete picture of the event.

Football Match Monitoring — Ullevaal Stadion

Owing to the isolated nature of the network infrastructure within the stadium — engineered to exclusively serve stadium attendees — it is feasible to closely correlate measured device counts with ticketing statistics. Mobile data also provides supplementary insights on supporter origins and overnight accommodation, making it an exceptionally valuable data source for event organisers and the Norwegian Football Federation (NFF).

Monitoring Area

Device counts were collected at approximately five-minute intervals from cell tower connections inside two nested geographic polygons covering the stadium and its immediate surroundings:

Area	Role
Ullevaal_Stadion	Stadium bowl polygon — finish-line accuracy zone covering the seating bowl, concourse and immediate ticketed perimeter.
Larger_Ullevaal_stadium_area	Wider polygon covering Sognsveien, the metro station, fan zones, and surrounding residential/commercial blocks.

Counts in this report use the stadium polygon unless otherwise stated. Baseline ambient population in the surrounding area is estimated at 1,500–2,500 Telenor devices ($\approx$ 3,000–5,000 people after extrapolation) at any given time outside match hours.

Event Calendar

Phase	Dates	Events / Context
Pre-Event Buffer	3 Sep 2025	Normal weekday baseline; pre-match operational checks.
Match 1	4 Sep 2025	Norway–Finland — International Friendly. Result: 1–0. Official attendance: 18,665.
Inter-Match	5–8 Sep 2025	Normal weekdays / weekend. Crowd recedes to baseline.
Match 2	9 Sep 2025	Norway–Moldova — FIFA World Cup Qualifier. Result: 11–1. Official attendance: 24,605.
October Block	1–10 Oct 2025	Normal baseline. NFF and police prepare extensive security operation for Israel match.
Match 3	11 Oct 2025	Norway–Israel — FIFA World Cup Qualifier. Result: 5–0. Official attendance: 19,363 (capacity restricted).
Post-Event	12–14 Oct 2025	Return to baseline; Norway–New Zealand friendly on 14 Oct (17,652 official).

Daily Device Overview

The chart below shows the average concurrent device count per day across the September and October monitoring windows. Match days are marked with red dashed lines.

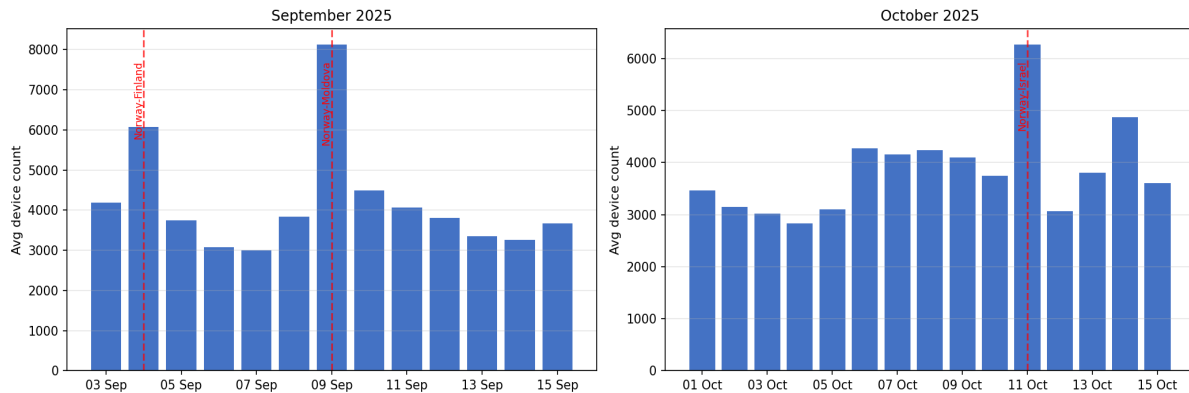


Fig. 1 — Daily mean device count at Ullevaal Stadion. The Moldova fixture (9 Sep) drives the highest daily mean of the September window; the Israel fixture (11 Oct) is the dominant peak in October.

Outside match windows, the polygon settles to a steady weekday-daytime baseline of approximately 4,200–4,500 devices. Match days lift the daily mean by 35–80% and produce sharp 2–4 hour spikes that fully utilise the stadium's 28,000-seat capacity.

Match-Day Crowd Patterns

All three matches share the characteristic stadium signature: a slow ramp-up two to three hours before kick-off as supporters arrive, a tight peak that holds for 90+ minutes during play, and a steep drop-off in the 30 minutes following the final whistle. The amplitude and shape of each spike, however, varies in revealing ways.

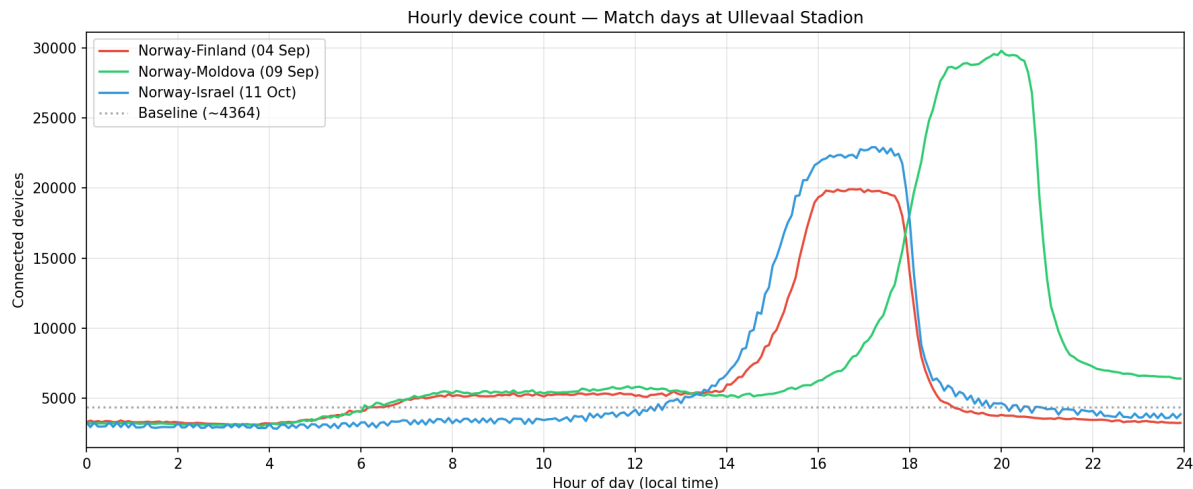


Fig. 2 — Hourly device counts on the three match days, plotted against the surrounding-area baseline (~4,400 devices). The Moldova fixture clearly mirrors a full-stadium attendance; Norway–Finland and Norway–Israel show comparable amplitudes despite very different competitive contexts.

Match	Date	Peak devices	Peak time	Official attendance
Norway–Finland (Friendly)	4 Sep 2025	19,938	16:55	18,665
Norway–Moldova (WC Qualifier)	9 Sep 2025	29,811	20:00	24,605
Norway–Israel (WC Qualifier)	11 Oct 2025	22,922	17:10	19,363

Peak counts include both ticketed spectators and the ambient population in the polygon (transit users, residents, visitors to nearby restaurants and the Norwegian Football Museum). Subtracting the ~3,000–5,000-person baseline brings the matchday excess into close agreement with official ticketing figures.

Norway–Finland | 4 September 2025 (International Friendly)

20,679 tickets sold

Telcofy peak: 19,938 devices @ 16:55 — fully aligned with the 18,665 official attendance plus venue-surrounding ambient population.

The Finland friendly served as the autumn opening fixture and as a low-stakes operational dress-rehearsal ahead of the World Cup qualifier four days later. Norway won 1–0; turnout was healthy for a friendly but well below the venue's 28,000 capacity. The early peak time (16:55, ahead of an 18:00 kick-off) reflects the typical Friday-evening stadium-pack pattern and was consistent with NFF gate-flow expectations.

Foreign-SIM presence was the highest of the three fixtures at 13.8%, led by Germany (45,028 readings), the Netherlands (39,401), Ukraine (17,875), and Poland (10,036). Despite being the visiting nation, Finnish SIMs registered only 5,918 readings on match day — indicating that the great majority of Finnish supporters travelled by ferry/road and likely roamed onto Norwegian networks before the dataset's filter window, or simply that the travelling Finnish support was numerically modest.

Norway–Moldova | 9 September 2025 (FIFA World Cup Qualifier)

~28,000 people at peak

Telcofy peak: 29,811 @ 20:00 — full stadium plus baseline. Confirmed against the 24,605 official attendance.

Norway–Moldova was the most attended fixture of the autumn cycle and the most attended match of the entire September–October monitoring window. Erling Haaland scored five and Thelonious Aasgaard added a hat-trick in an 11–1 win that has already entered Norwegian football folklore. The polygon's 29,811 peak at 20:00 confirms that the venue effectively reached its 28,000 working capacity, with the residual delta absorbed by the surrounding-area baseline.

Foreign-SIM share was the lowest of the three at 10.6% — a typical pattern for a ticket-rich domestic-friendly opponent: Norwegian fans heavily out-numbered the small Moldovan travelling support (no MD-flagged SIMs were detected in the daily breakdown, consistent with Moldova's small national-team supporter footprint). German (48,835), Dutch (42,337) and Ukrainian (15,849) SIMs again dominated the international tail.

Norway–Israel | 11 October 2025 (FIFA World Cup Qualifier)

~23,000 people on site

Telcofy peak: 22,922 @ 17:10. Excluding the 3,000–5,000 baseline, this is consistent with the 19,363 official attendance under restricted capacity.

The 11 October qualifier against Israel was Ullevaal's most operationally complex fixture in modern memory. Norwegian authorities mounted what they described as the largest security operation at a sporting event since the 1994 Lillehammer Olympics — including a no-fly zone over Sognsveien, reinforced anti-terror cordons and stringent body searches at every entry point. Sections of the bowl near the visiting fan area were closed to create a security buffer, reducing the venue's effective capacity from ~28,000 to a working ~20,000.

Inside the stadium polygon, Telcofy's peak of 22,922 devices is fully consistent with the 19,363 ticketed attendance plus the surrounding ambient population and event-day staff/security overhead. The matchday peak time (17:10) is markedly earlier than the Moldova peak (20:00), reflecting the policy decision to admit ticket-holders earlier than usual to prevent congestion at security checkpoints.

Telcofy additionally monitored supplementary geofenced zones in central Oslo where pro-Palestinian demonstrations took place during the day, providing the federation with a city-wide view of crowd distribution. Between 1,000 and 1,500 protesters marched from Spikersuppa Square to the stadium perimeter; Israeli SIMs in the Ullevaal polygon registered only 2,554 readings throughout match-day, indicating an extremely small travelling Israeli support.

“19,000 saw the match between Norway and Israel. I expect something like that — 1–2,000 people in demonstration and the rest are people in the area who work, live at a hotel, visit restaurants, etc. The total of 23,000 we see from adding the extrapolation is acceptable.”

— Norwegian Football Federation

International Visitor Analysis

Using Mobile Country Code (MCC) data from device SIM registrations, Telcofy estimates the share of Norwegian versus international devices inside the Ullevaal Stadion polygon on each match day. Note that REST aggregates SIMs from operators that do not transmit a usable country code; it is not a true "country" but rather an unclassified-roaming bucket.

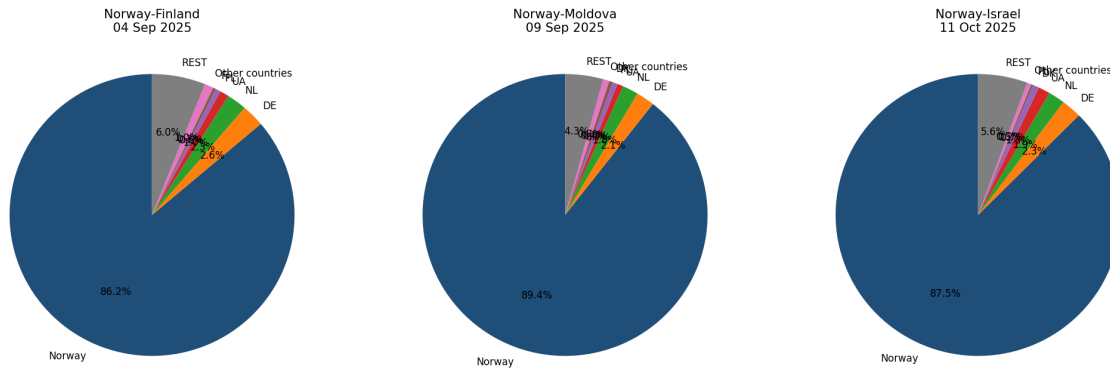


Fig. 3 — Country composition of devices inside the Ullevaal Stadion polygon on each of the three match days. Norwegian devices dominate at 86–89%; the foreign tail is led by Germany, the Netherlands and Ukraine on every match day.

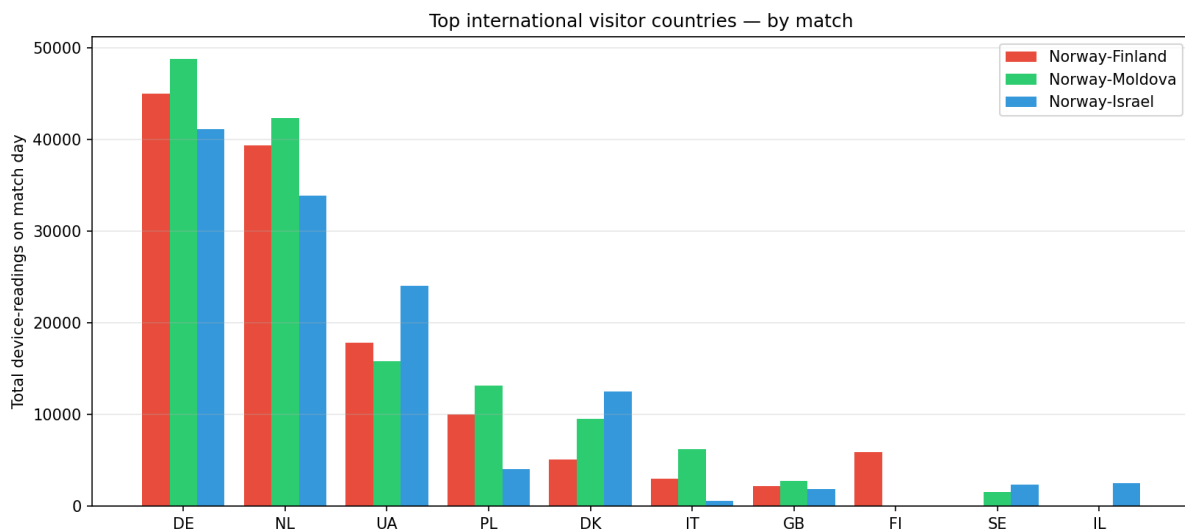


Fig. 4 — Top international visitor countries by match day. Germany, the Netherlands and Ukraine appear on every podium; Denmark replaces Poland as a top-five source for the Norway–Israel fixture.

Top non-Norwegian source markets (combined match-day device-readings)

Country	Norway–Finland	Norway–Moldova	Norway–Israel	Context
Germany (DE)	45,028	48,835	41,154	Largest non-REST foreign cohort across all three matches; reflects German football tourism and resident expat

Country	Norway–Finland	Norway–Moldova	Norway–Israel	Context
				population in Oslo.
Netherlands (NL)	39,401	42,337	33,910	Consistent #2; Dutch expat community and football tourism.
Ukraine (UA)	17,875	15,849	24,041	Surge for the Israel match — reflects Norway's large resident Ukrainian community.
Poland (PL)	10,036	13,137	4,102	Strong September presence; ebbs by October.
Denmark (DK)	5,125	9,534	12,527	Steadily increases match by match — neighbour-country support.
Finland (FI)	5,918	—	—	Visiting-team SIMs only on the Finland match-day.
Israel (IL)	—	—	2,554	Very small travelling-supporter footprint, consistent with security advisories.
REST (unclassified)	105,834	101,049	101,064	Roaming SIMs without a transmitted MCC —

Country	Norway–Finland	Norway–Moldova	Norway–Israel	Context
				includes a portion of long-haul international visitors.

Key Takeaways

- **Moldova was the most attended fixture of the autumn window:** the 9 September qualifier produced the highest concurrent peak (29,811 devices) of the September–October monitoring period, fully consistent with the 24,605 official attendance plus surrounding-area baseline. The Haaland five-goal performance produced a true full-house response to a nominally low-prestige opponent.
- **Stadium polygon counts correlate tightly with ticketing data:** after subtracting the 3,000–5,000-person ambient baseline, all three peaks reproduce the official attendance figures within a few percent — validating Telcofy's network-signal methodology as a real-time proxy for ticketed attendance.
- **The Israel fixture was attendance-restricted, not unpopular:** the 22,922 polygon peak versus the 19,363 official attendance reflects the deliberate capacity restriction (security buffer in sections nearest the visiting fan area), not weak demand. Crowd-arrival patterns shifted earlier in the day, consistent with the enhanced security-check throughput.
- **Foreign-SIM mix is dominated by European resident populations and tourism, not visiting-team supporters:** Germany, the Netherlands, Ukraine, Denmark and Poland accounted for the majority of identified foreign devices on every match day. Pure visiting-team support (Finland, Moldova, Israel) registered far smaller footprints — the international colour at Ullevaal is structural, not match-driven.
- **Tøyenparken needed temporary infrastructure; Ullevaal did not:** the open-air festival was only monitorable thanks to Telenor's temporary antenna deployment, and the data outage on 6 August was caught in real time via abnormally low device counts. Ullevaal's permanent in-stadium cellular fabric handled all three matches without operational anomalies.

Methodology

Device counts are derived from anonymised and aggregated mobile-network signals from Telenor, the largest Norwegian operator. Counts represent concurrent connected devices inside a defined geographic polygon, sampled approximately every five minutes. International visitor estimates use the Mobile Country Code (MCC) of the device's SIM registration; the "REST" bucket aggregates roaming SIMs that do not transmit a usable MCC.

Where extrapolated headcounts are quoted (e.g. for the open-air festival), Telcofy applies a 45% Telenor market-share factor to convert observed devices into a total-people estimate. For the Ullevaal stadium polygon the device counts in this report already reflect total concurrent presence at the polygon level, calibrated against ticketing ground-truth.

All processing is carried out in compliance with applicable privacy regulations (GDPR / Norwegian personal data law). No individual identifiers are exposed at any stage; all reporting is at aggregate polygon level only.